📋 AI LOAN UNDERWRITING PLATFORM - QUICK REFERENCE GUIDE

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🏗️ ARCHITECTURE OVERVIEW

Frontend (What You See)
├─ HTML5 (Structure)
├─ CSS3 (Design & Animations)
├─ JavaScript (Logic & Calculations)
├─ Chart.js (Visualizations)
├─ Font Awesome (Icons)
└─ XLSX (Excel Export)

Data Flow (How It Works)
Upload → Extract → Validate → Process → Calculate → Score → Recommend → Export

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💻 TECHNOLOGIES USED

TECHNOLOGY	VERSION	PURPOSE	DEPENDENCY
HTML5	5	Structure & Forms	None
CSS3	3	Styling & Animations	None
JavaScript	ES6+	Core Logic	None
Chart.js	3.9.1	Data Visualization	CDN
XLSX (SheetJS)	0.18.5	Excel Generation	CDN
Font Awesome	6.4.0	Icon Library	CDN
Google Fonts	Latest	Typography	CDN

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📊 CORE CALCULATIONS & FORMULAS

1. EMI Calculation (Equated Monthly Installment)
   Formula: EMI = [P × R × (1+R)^n] / [(1+R)^n - 1]
   Where:
   - P = Principal Loan Amount
   - R = Monthly Interest Rate (Annual Rate ÷ 12)
   - n = Number of Months

   Example: ₹10,00,000 @ 10% p.a. for 60 months = ₹21,247

2. FOIR - Fixed Obligation Income Ratio
   Formula: FOIR = (Total Monthly Obligations ÷ Monthly Income) × 100
   Safe Range: < 40%
   Acceptable: 40-55%
   Risk: 55-70%

   Example: EMI ₹21,247 + Existing ₹15,000 = ₹36,247 ÷ ₹75,000 = 48.3%

3. DSCR - Debt Service Coverage Ratio
   Formula: DSCR = (Monthly Income - Total Obligations) ÷ Proposed EMI
   Healthy: ⩾ 1.25x
   Marginal: 1.0-1.25x
   Stressed: < 1.0x

   Example: (₹75,000 - ₹36,247) ÷ ₹21,247 = 1.82x (Healthy)

4. PTI - Payment-to-Income Ratio
   Formula: PTI = (Proposed EMI ÷ Monthly Income) × 100
   Safe: < 30%
   Moderate: 30-45%
   Risk: > 45%

5. LTI - Loan-to-Income Ratio
   Formula: LTI = Loan Amount ÷ Annual Income
   Conservative: < 2.0x
   Moderate: 2.0-3.5x
   Aggressive: > 3.5x

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🎯 RISK SCORING ALGORITHM

Creditworthiness Score (0-100):

  Base: 50 points

  Add:
  + (CIBIL Score ÷ 900) × 20   [0-20 points]
  + FOIR Assessment   [5-20 points] (Lower FOIR = More points)
  + DSCR Assessment   [0-15 points] (Higher DSCR = More points)
  + Employment Stability   [3-10 points] (Salaried = 10, Gig = 3)
  + Liquidity Buffer   [2-10 points] (Strong AMB = 10)
  + Current Ratio   [3-8 points]   (Higher CR = More points)
  + Bounce History   [0-7 points]   (Clean = 7, Frequent = 0)

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  + Work Orders Pipeline          [0-5 points]   (Strong = 5)

  Final Score = Sum (capped at 100)

Risk Tiers:
  75-100 → Tier 1: APPROVE ✅ (Low Risk)
  55-74  → Tier 2: CONDITIONAL ⚠️ (Medium Risk)
  35-54  → Tier 3: MANUAL REVIEW 🔍 (High Risk)
  0-34   → Tier 4: DECLINE ❌ (Critical Risk)

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📦 CORE FEATURES

Data Extraction:
- Drag-and-drop file upload
- Pattern matching for field extraction
- Multi-format support (PDF, Excel, Images)
- Auto-population of forms

Financial Analysis:
- Income verification
- Liquidity assessment
- Profitability analysis
- Cash flow evaluation

Credit Assessment:
- CIBIL score evaluation
- Bureau data integration
- Bounce detection
- Historical analysis

Ratio Analysis:
- FOIR (Fixed Obligation Income Ratio)
- DSCR (Debt Service Coverage Ratio)
- PTI (Payment-to-Income Ratio)
- LTI (Loan-to-Income Ratio)
- Current Ratio
- Debt-to-Equity Ratio
- Profit Margin

Risk Assessment:
- Weighted scoring model
- Multi-factor analysis
- Fraud detection flags

- Trend analysis

Decision Support:
- AI-generated recommendations
- Structured decision output
- Risk tier classification
- Suggested loan structure

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🎨 UI/UX ELEMENTS

Colors Used:
  Primary: #0F172A (Dark Navy) - Headers, Important elements
  Secondary: #EF4444 (Red)   - Alerts, Call-to-action
  Accent: #1E40AF (Blue)   - Links, Focus states
  Success: #10B981 (Green)   - Positive indicators
  Warning: #F59E0B (Amber)   - Cautions
  Danger: #EF5350 (Red)   - Errors
  Light: #F9FAFB (Off-white) - Backgrounds

Typography:
  Font Family: Segoe UI, Tahoma, Geneva, Verdana, sans-serif
  Headings: Bold, larger sizes
  Body: Regular, 14-16px
  Monospace: Data values, numbers

Animations:
  Duration: 0.3 seconds (standard)
  Easing: ease, ease-in-out
  Effects: Fade-in, slide, scale, color-change

Layout:
  Grid System: CSS Grid, Flexbox
  Responsive: Mobile-first, breakpoints at 768px
  Spacing: 20px base unit

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📈 VISUALIZATION TYPES

Chart.js Integration:
  ✓ Doughnut Charts (Application Status Distribution)
  ✓ Bar Charts   (Risk Levels, Segment Analysis)
  ✓ Line Charts   (Trends, Timeline)

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  ✓ Pie Charts       (Portfolio Composition)

Custom Visualizations:
  ✓ Progress Bars    (Ratio tracking)
  ✓ Status Badges    (Color-coded status)
  ✓ Heat Maps        (Risk distribution)
  ✓ Metric Cards     (KPI display)
  ✓ Summary Tables   (Data tables)

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🔧 FORM INPUTS & FIELDS

Personal Information:
  - Name, Email, Phone, Age, Pan, Aadhar

Employment Details:
  - Employment Type (Dropdown)
  - Company/Business Name
  - Designation
  - Work Experience (Years)

Income Information:
  - Monthly Income (₹)
  - Annual Income (₹)
  - Income Stability Score

Loan Details:
  - Loan Amount (₹)
  - Tenure (Months)
  - Existing EMI (₹)
  - Proposed Use

Financial Metrics:
  - Bank Name
  - Avg Monthly Balance
  - Turnover
  - Net Profit
  - Cash Flow
  - Current Ratio
  - Debt-to-Equity

Business Information:
  - GST Registration (Yes/No)
  - Work Orders Count
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  - Work Orders Value
  - CIBIL Score

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📊 BORROWER ATTRIBUTES (Complete List)

Personal:
  name, email, phone, age, pan, aadhar, gender, address

Employment:
  type, company_name, designation, experience_years, tenure_months, salary_type

Income:
  monthly_income, annual_income, salary_slip, bonus, allowances, gross_income

Banking:
  primary_bank, account_type, account_age_months, avg_monthly_balance,
  monthly_turnover, bounce_count, savings_balance, credit_balance

Credit:
  cibil_score, existing_emi, total_liabilities, credit_history_age,
  number_of_loans, number_of_credit_cards, defaults_count

Financial:
  turnover, net_profit, revenue, expenses, profit_margin,
  current_assets, current_liabilities, fixed_assets

Business:
  gst_registration, gst_number, work_orders_count, work_orders_value,
  business_age_years, industry_type, customer_concentration

Loan Request:
  loan_amount, tenure_months, purpose, collateral_type, collateral_value,
  co_borrower, guarantor

Calculated:
  proposed_emi, total_obligations, foir, dscr, pti, lti,
  creditworthiness_score, risk_tier, recommendation

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🔐 SECURITY & COMPLIANCE

Data Protection:
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✓ Client-side processing (no server needed)
✓ Input validation & sanitization
✓ No sensitive data in console logs
✓ Optional LocalStorage encryption

Recommended Backend Security:
- HTTPS/TLS encryption
- AES-256 data encryption
- JWT authentication
- Rate limiting
- RBAC (Role-Based Access Control)
- Audit logging
- Two-factor authentication

Compliance Standards:
- RBI Fair Practices Code
- Know Your Customer (KYC)
- Anti-Money Laundering (AML)
- DPDPA (Data Protection)
- GDPR (if international)

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📱 RESPONSIVE DESIGN

Desktop: Full layout with all features
- Multi-column grids
- Large charts and tables
- Side-by-side panels

Tablet: Optimized layout
- 2-column grids
- Medium-sized charts
- Adjusted spacing

Mobile: Single-column layout
- Vertical stacking
- Full-width inputs
- Touch-friendly buttons
- Collapsible navigation

Breakpoints:
- 1200px: Desktop
- 768px: Tablet
- 480px: Mobile

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⚡ PERFORMANCE METRICS

Load Time:
  • First Load: < 2 seconds
  • Page Switch: < 300ms
  • Chart Rendering: < 500ms
  • Excel Generation: < 2 seconds

File Sizes:
  • Standard Version: ~500 KB
  • Pro Version: ~750 KB
  • All-in-one HTML (no dependencies)

Optimizations:
  • Minified CSS
  • Optimized JavaScript
  • CDN-based libraries
  • Lazy loading for charts
  • Efficient DOM manipulation

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🚀 DEPLOYMENT OPTIONS

Option 1: Local (Offline)
  • Save HTML file on computer
  • Open in browser
  • Works immediately, no internet needed

Option 2: Local Server
  • Python: python -m http.server 8000
  • Node: npx http-server
  • Port: http://localhost:8000

Option 3: Cloud Hosting
  • GitHub Pages (Free, static)
  • Netlify (Free, serverless)
  • Vercel (Free, serverless)
  • AWS (Scalable, production)
  • Google Cloud (Enterprise)
  • Azure (Enterprise)

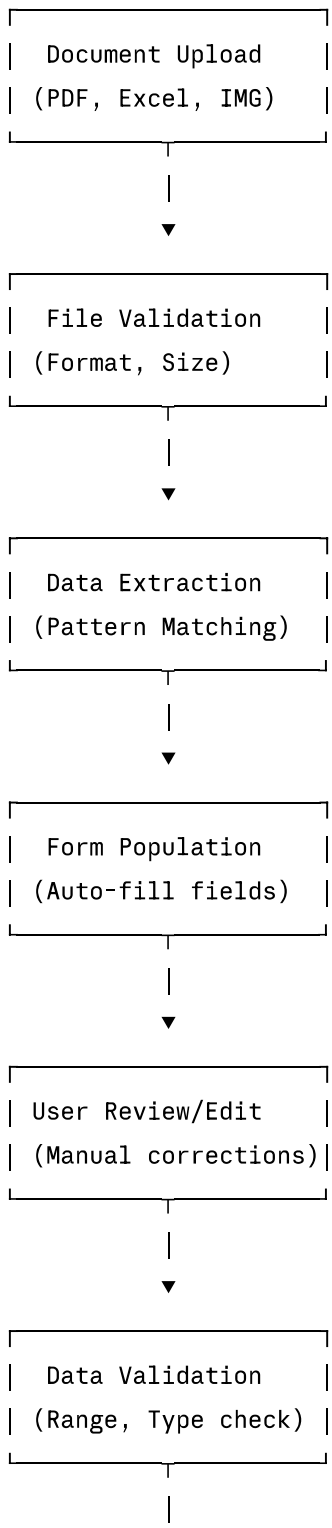
```
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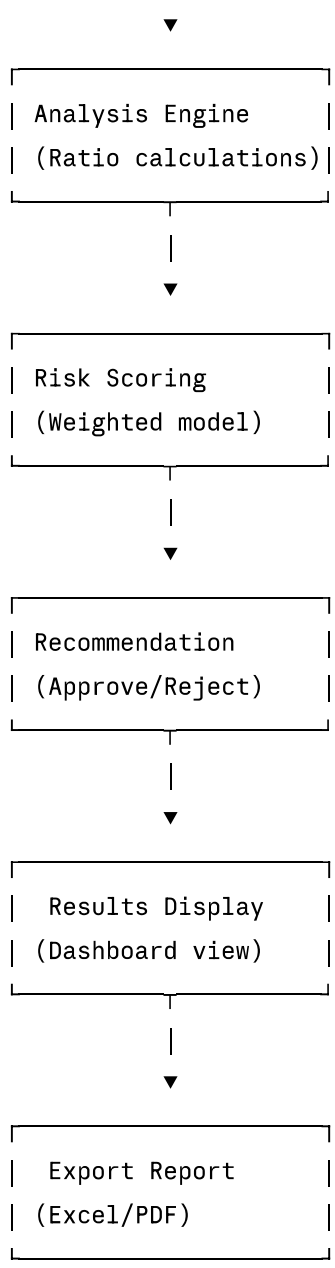
Option 4: Corporate Network

- Internal server deployment
- Database integration
- User authentication
- Full enterprise setup

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📊 DATA FLOW DIAGRAM





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📋 BROWSER COMPATIBILITY

Desktop Browsers:
  ✓ Chrome 90+
  ✓ Firefox 88+
  ✓ Safari 14+
  ✓ Edge 90+

Mobile Browsers:
  ✓ Chrome Android 90+
  ✓ Safari iOS 14+
  ✓ Firefox Android 88+
  ✓ Samsung Internet 14+

Requirements:

- JavaScript enabled
- ES6 support
- CSS Grid & Flexbox support
- File API support (for upload)
- LocalStorage support (optional)

💡 KEY ADVANTAGES

✅ No Backend Required
- Works completely standalone
- No server dependency
- Can be deployed offline

✅ Fast Performance
- Client-side processing
- Instant calculations
- No network latency

✅ Secure
- Data stays on user's device
- No cloud transmission
- GDPR/DPDPA compliant option

✅ Easy to Deploy
- Single HTML file
- No installation required
- Works in any browser

✅ Scalable
- Add backend API when needed
- Integrate with external services
- Deploy to enterprise systems

✅ Professional
- Enterprise-grade UI/UX
- Advanced analytics
- Complete feature set

🎯 USE CASES

1. NBFC Loan Processing
   - Automate document analysis
   - Speed up credit assessment
   - Reduce manual work
   - Improve consistency

2. Bank Internal Assessment
   - Pre-screening applications
   - Portfolio risk analysis
   - Compliance reporting
   - Trend monitoring

3. Credit Consultant Tools
   - Client financial analysis
   - Loan affordability calculator
   - Investment planning
   - Risk assessment reports

4. Fintech Applications
   - Instant loan approval system
   - Embedded lending
   - Credit scoring
   - Portfolio management

5. Training & Education
   - Credit analyst training
   - Risk management courses
   - Financial literacy
   - Demonstration platform

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📞 SUPPORT & DOCUMENTATION

Documentation:
- Technical Architecture Document
- API Integration Guide
- Database Schema
- Frontend Code Comments
- Deployment Guide

Resources:
- Chart.js Documentation: https://www.chartjs.org
- XLSX Documentation: https://github.com/SheetJS/sheetjs

- Font Awesome Icons: https://fontawesome.com
- HTML5 Reference: https://developer.mozilla.org/en-US/docs/Web/HTML
- CSS3 Reference: https://developer.mozilla.org/en-US/docs/Web/CSS
- JavaScript Reference: https://developer.mozilla.org/en-US/docs/Web/JavaScript

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✨ NEXT STEPS FOR ENHANCEMENT

Phase 2 (Backend Integration):
- Add database (PostgreSQL/MongoDB)
- Implement authentication
- Create REST API endpoints
- Add user management
- Audit trail logging

Phase 3 (Advanced Features):
- Real CIBIL/GST API integration
- OCR document processing
- Machine learning models
- Predictive analytics
- Automated collections

Phase 4 (Enterprise):
- Multi-user support
- Role-based access
- Workflow automation
- Custom report builder
- Mobile native apps

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VERSION HISTORY

v1.0 (Current)
  ✓ Basic loan analysis
  ✓ Standard form & calculations
  ✓ Excel export
  ✓ Single-page layout

v1.0 Pro
  ✓ Advanced dashboard
  ✓ Portfolio management
  ✓ Multiple pages
  ✓ Advanced analytics

✓ Compliance tools
✓ Professional UI

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CONTACT & FEEDBACK

For questions, feedback, or customizations:

- Review the technical documentation
- Check browser console for errors
- Verify data input formats
- Test with sample data first
- Contact support for enterprise deployment

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Last Updated: January 2025
Platform Version: 1.0 (Standard & Pro)
Status: Production-Ready ✓
License: Ready for deployment

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